

From Activity-Based-Model to MATSim for BRT Planning: Practical Lessons from the Tel Aviv Pink Route

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Introduction

Public transit (PT) planning teams often lack access to a reliable and reasonably fast way to assess the service or network changes effect on the riders. Partly, because the transportation models are quite complex and expensive to maintain and thus require a separate team or consultant, and the model configurations and runs take much longer, than would be reasonable for the typical planning cycle; partly, because the typical TAZ-based assignment model lacks the necessary level of granularity and temporal resolution to analyze changes in the transit schedule, passenger behavior or transfer comfort and reliability. As a result, often, only the rail-based modes and trams get modelled, while bus networks are mostly only simulated during the strategic planning phases once in several years, when significant network updates are getting approved.

On the other hand, transit planners are actively using GIS tools with simple heuristics and calculations like isochrone analysis or the population and workplace assessment for the line or stop catchment areas. While Replan PT planning suit provides extensive GIS tools to work with the PT networks and schedules (GTFS-based), or ambition in the this project was to empower the public transit authorities to use the more suitable for the PT assignment agent-based simulation in MATSim to plan and operate bus networks and quickly assess sketched solutions.

1. Converting an existing activity-based model

Transport authorities rarely start a modelling process from a blank sheet. Normally, they already have an approved strategic model (maintained either internally or by contractors), established forecast procedures and years of accumulated assumptions that their planning process depends on. The practical question is therefore often not how to build a MATSim model, but how to add agent-based simulation without replacing the model that is already trusted and used for forecasting.

We encountered this problem in a 2026 project for Netivei Ayalon around the planned Pink BRT corridor connecting Herzliya, Ra'anana and Kfar Saba. The client already had an activity-based metropolitan EMME demand model. Our task was to convert its demand into a MATSim simulation, calibrate the resulting model against observed public transport use, and make it usable for comparing the BRT and surrounding public transport network alternatives in replan.city.

This paper reports what proved difficult, what was transferable, and what we would do differently when converting another strategic model.

At first sight the source data looked unusually well suited to MATSim. The EMME synthetic population contains person and household attributes, primary and secondary tours, intermediate stops, activity purposes, destination zones and a main travel mode. We therefore did not have to reconstruct daily activity chains from aggregate OD matrices.

However, converting these records into MATSim plans was not simply a file-format transformation. The strategic model describes some decisions at a different level of detail. Time is encoded in broad periods rather than exact departure times, and locations as traffic analysis zones rather than coordinates. A tour has a main mode, while MATSim needs a sequence of individual legs. Combined modes such as park-and-ride have to be expanded into access, public transport and egress legs, and mode changes on the return part of a tour have to be preserved. Driver and car-passenger trips also have different meanings.

These gaps required assumptions. In practice, conversion became a process of interpreting the source model first and only then translating it into MATSim. This meant revisiting the EMME documentation, checking field semantics with the model developers, and adding temporal and spatial disaggregation where the strategic model did not contain the detail required by MATSim (cf. Ben-Dor et al., 2021).

There was also a temporal mismatch. The demand structure comes from the calibrated 2018 strategic model, while the road network, GTFS supply, bus-lane inventory and Rav-Kav observations represent 2026 conditions. The resulting base is therefore a hybrid rather than a pure 2026 demand model. This is acceptable for the corridor comparison, but needs to be kept in mind when interpreting absolute demand.

2. Calibration and transferability

Calibration is only summarized here because the scope of publication of the detailed sequence is still being reviewed with the client. The main lesson was that microscopic PT simulation is very sensitive to how the transport supply is represented. Network QA, including lane information and dedicated bus infrastructure, had to be resolved before behavioural parameters could be interpreted sensibly. Aggregate ridership alone was also not enough: walking, waiting, transfers and in-vehicle time were checked because these are precisely the mechanisms through which a BRT scenario affects passengers. The final base-year calibration also included a line-level adjustment to observed passenger counts, following the general logic of count-based public transport calibration in MATSim (Moyo Oliveros and Nagel, 2016).

For forecasting, however, it is important to separate the parts of the model that can be transferred from those that are specific to the observed base year. Demand remains the responsibility of the strategic model. A future-year synthetic population can be passed through the same conversion procedure, which assigns exact times and locations and translates the strategic tour description into MATSim plans. The MATSim layer then adds the road and PT network representation, routing and replanning logic, and the behavioural parameters calibrated in the base model. Planned infrastructure and public transport services change between scenarios, but the underlying simulation and behavioural setup should normally remain unchanged. Adjustments fitted directly to observed base-year services are different: they are specific to the network against which they were calibrated.

This distinction is particularly important for line-level passenger counts. A count correction can be useful for reproducing the existing network, but a future BRT network will have different routes, headways and transfer opportunities. Carrying the old line allocation into such a scenario would partly prevent the passenger redistribution that MATSim is supposed to simulate. We therefore also tested the calibrated model after releasing agents back into normal replanning, without preserving the line-specific allocation. The exact line-by-line fit changed, while aggregate and structural correspondence with the observed system remained much more stable. We treat this relaxed run as a test of what survives when the model is used for scenario analysis, rather than as another calibration target.

The forecast workflow follows directly from this distinction. Forecast demand is taken from the strategic model and converted using the same rules as in the base year. Forecast road and public transport supply are then built on top of the calibrated base representation. Behavioural and replanning parameters are retained, while base-year line-count corrections are removed. Scenarios are compared using the same set of indicators. This also defines the level at which the results should be interpreted: corridor and trunk demand, mode shift, accessibility and the relative performance of alternatives are more transferable than the exact patronage of an individual low-volume feeder line.

3. Transferability

Not all parts of the calibrated base model should be carried into future scenarios. Forecast demand still comes from the strategic model; MATSim provides the reusable simulation layer.

Component	Transferable to future scenarios?	Treatment
Synthetic population and activity chains	No	Replace with future-year demand from the strategic model
Time and location disaggregation	Yes	Apply the same conversion rules

Component	Transferable to future scenarios?	Treatment
Corrected road and PT representation	Yes	Keep as base, adding planned infrastructure and services
Behavioural and replanning parameters	Yes	Retain unless the scenario requires a different assumption
Line-specific count correction	No	Remove, because future routes and passenger allocation change

We tested this by releasing the calibrated population back into normal replanning without preserving the line-specific allocation. The exact line fit changed, but the overall structure remained stable: total PT boardings were within 6.3% of observations, passenger-weighted error was 15.7%, and simulated and observed line volumes had $R^2 = 0.976$.

We therefore interpret future scenarios mainly at corridor and trunk level, while individual low-volume feeder lines carry greater uncertainty.

4. What planners can use

For planners, the useful output is not the MATSim event stream itself but the ability to interrogate a scenario. In *replan.city* the same run can be inspected by line, segment, stop and time period, together with waiting and transfer patterns, travel-time components, accessibility, PT-versus-car competitiveness and operating quantities such as vehicle-kilometres and vehicle-hours. The same views can be used for the baseline, the BRT with the existing network and redesigned feeder alternatives.

The first client workshop also showed the value of keeping this interaction simple. Participants immediately used filters to find problematic stops and changed the time period to inspect how patterns moved through the day. Modelling expertise is still needed for conversion, network QA and uncertainty; the difference is that scenario results no longer have to remain inside the modelling team.

5. Practical conclusions

The project left us with three points worth carrying into the next conversion. An established strategic model has to be interpreted, not merely imported, because MATSim needs detail that the source model may not contain explicitly. Calibration of an agent-based PT model cannot be separated from network and supply QA. And for future scenarios, the useful object to transfer is not the exact base-year line fit but the demand-conversion and behavioural simulation framework that continues to work when agents are allowed to adapt. This supports comparative corridor and trunk analysis more strongly than precise forecasts for individual low-volume services.

6. Presentation

At the MATSim User Meeting we will show the conversion workflow, the base-year calibration at a high level, the transferability test and examples of scenario analysis in *replan.city*.

References

- Ben-Dor, G., Ben-Elia, E., Benenson, I. (2021). Population Downscaling in Multi-Agent Transportation Simulations. Simulation Modelling Practice and Theory, Elsevier.
- Horni, A., Nagel, K., Axhausen, K. W. (eds.) (2016). The Multi-Agent Transport Simulation MATSim. Ubiquity Press, London.
- Moyo Oliveros, M., Nagel, K. (2016). Automatic Calibration of Agent-Based Public Transit Assignment Path Choice to Count Data. Transportation Research Part C, Elsevier.